

Lecture 7

IRM:
$$Loss_{IRM} = \sum_{i=1}^S L_{CE_i} + \lambda \sum_{i=1}^S \left\| \nabla L \right\|_2 \bigg|_{w=w^*}$$

Regulariser Strength $\rightarrow$ penalty for overfitting

SAM:

a little bit disturbance in Θ will move loss a lot more (change in loss) is more in Θ sharp

$\downarrow$
 $\Theta \pm \underline{\epsilon}$, change in loss is acceptable?
 $\downarrow$
 epsilon, perturbation

$$\therefore \epsilon \propto \Delta L$$

$\rightarrow$ means checking gradient steep or not $\nearrow$ if steep, ϵ is less
 $\rightarrow$ Hessian to see curvature

Law of Large numbers — bias decrease

— Shape — global causal feature
 └ never destroy it

→ Hard augmentation example

 └ Change texture of a car but label
 is same

— LCE ~~word~~ doesn't work on latent

— Bringing both together using MSE, contrastive

* Feature ~~end~~ disentanglement:

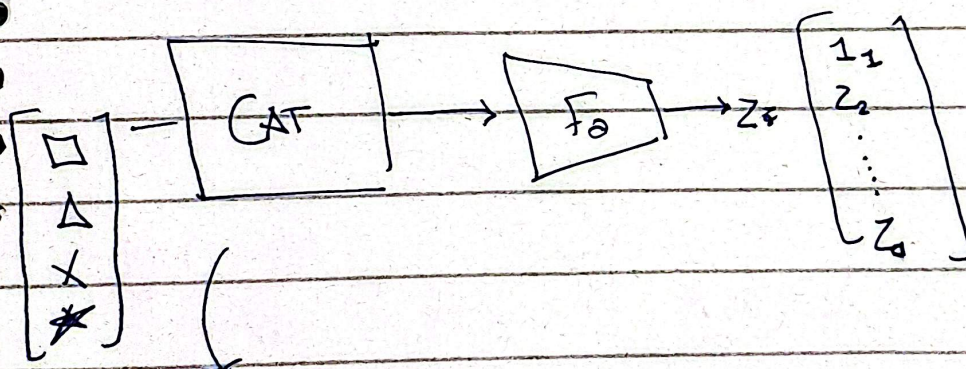


image car's image is being generated because of

Generative Factors where each Factor respond

some Feature of car like $\square$ — shape
 Δ — texture
 X — color

— We want Feature space to distinguish each generative
 Features → which disentangled

L_{CE} is greedy and there is high chance that features

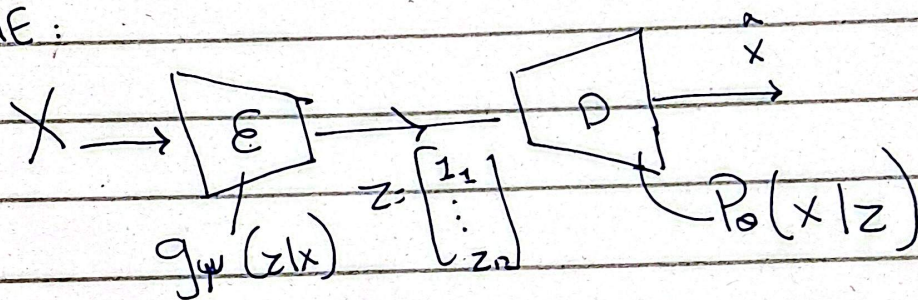
are entangled

↳ Also, we don't know how many generative factors are there?

↳ even if we know, how to disentangle them

Goal: Each dimension has a separate generative factor

VAE:



$$L_{\text{VAE}} = \text{MSE}(x, \hat{x}) + \underbrace{D_{\text{KL}}(q_{\phi}(z|x) || P(z))}_{\substack{\text{Produces} \\ \text{blurry images}}}$$

$N(0, I)$
isotropic gaussian

→ As B began to increase,

they become isotropic gaussian which is identical.

* Home project.

Human Face
dataset

Choose β
as hyperparameter

Fix $d-1$ dimension
and change last dimension

→ We want β to not make this 0, so this
doesn't cause posterior collapse

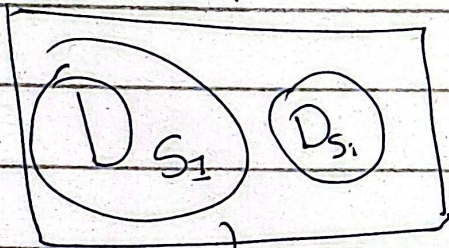
- Features only to be causal

↳ z_1 be independent of z_2

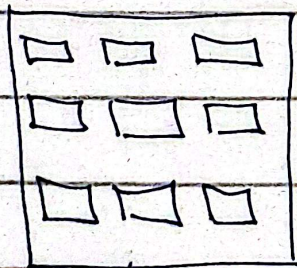
↳ also have information
about label

domain generalised

- Self Supervised Learning (SSL):

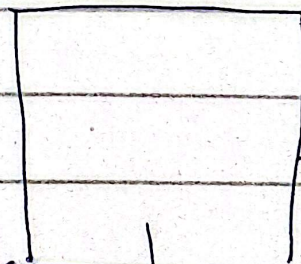


$\{x_i, y_i\} \rightarrow \{x_i\}$



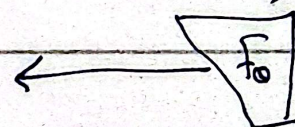
9 Patches of an image

Shuffle



100 images
even though 9!

unshuffle



→ Only unshuffle, if the image's mapping has been understood

↳ Done by classification →

↳ class classification



Pretext task → Finetune them

Dino

Q. How does this help Domain Generalise

Open Set Recognition [OSR]:

X : input space

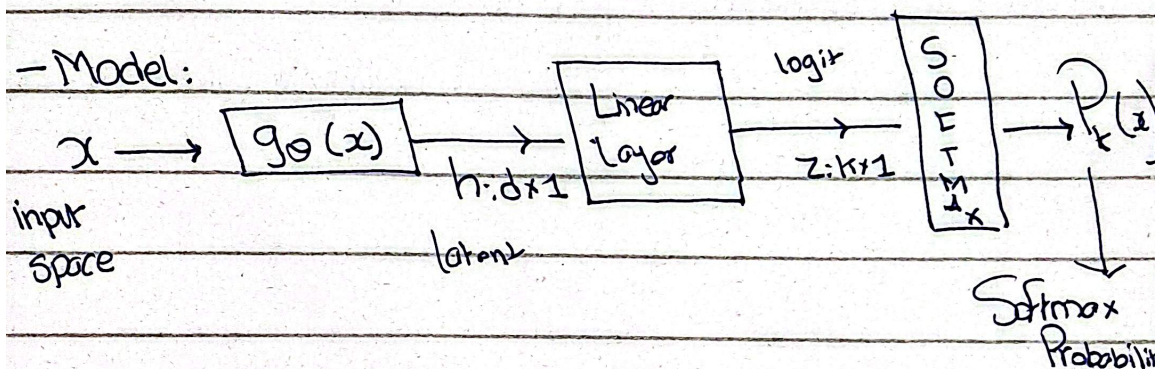
y_k : known classes $\{1, 2, \dots, k\}$ $\rightarrow$ closed set

y_u : unknown classes $\{\dots\}$ $\rightarrow$ open set

$$y_k \cap y_u = \emptyset$$

$P_{y_k} \neq P_{y_u}$ $\rightarrow$ Assumption

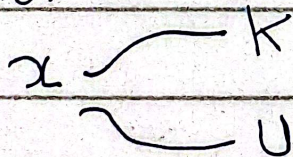
- Model:



Goal:

1. classifier — closed set

2. detector —



Single class classification

Anomaly / Outlier detector:

Out of distribution: $\rightarrow$ input from training distribution or not

$\hookrightarrow$ model training on cat pointing

$\hookrightarrow$ cat's real image from that distribution or not

Open set Recognition: y_k, y_u

$x = \begin{cases} y_k \\ y_u \end{cases}$ $\xrightarrow{\text{Classify it correctly}}$

Open world Recognition:

f_0 was trained on y_k

$x \notin y_k \rightarrow \boxed{f_0} \rightarrow y$

Classify unknown
one by putting

make one
more class

detect and see
it isn't y_k

then ~~the~~ $K+1$ classes

$\hookrightarrow$ 1. Learn those classes

2. Increment #classes

3. train model on that

4. modify weights

$\rightarrow$ Continued Learning
 $\downarrow$
LLM

$U(x)$ = unknownness Score

↓

how much unknown that image is for the model

↳ $\leq \tau \Rightarrow x \in y_k$

↳ $\geq \tau \Rightarrow y \in y_u$

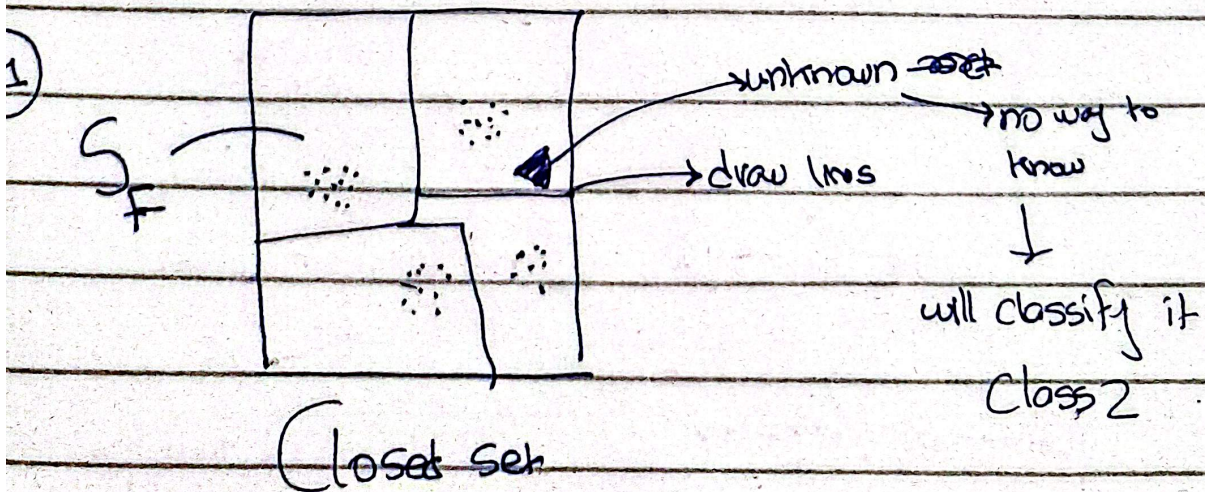
Types of Unknowns

↳ Near U — close to feature space
(difficult to reject)

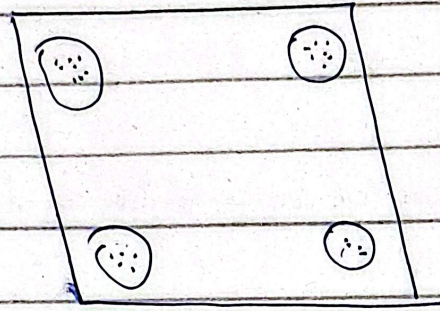
↳ Far U — Far to Feature space

Near U — Cat and leopard

↳ their Feature's space are closer



Open Space Risk



- Open set model
- classes occupy specific region

— holder note

— Open Space Risk in Open Set is more than maximum

Q. How to find where the unknowns will lie?

Find optimal unknown

$$U(x) = \log \left(\frac{P_u(x)}{P_k(x)} \right)$$

assuming where it will land

the larger $U(x)$, the more confident model

Likelihood Ratio test